EMCH 502: Engineering Analysis

Final Exam Take Home | Spring 2026

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Please see Attached 4 Data File from a Milling Machine. These four files were collected at four different times.

Problem 1: (Both UG and Grad)

We need to predict if file names FOC_20190801_06.csv and FOC_20190812_09.csv indicate any abnormal behavior or not. Bianary - 'Yes' or 'No'

There is a column in each files called - MACHINE ALARM - which if has a code [070008], indicates the machine was at fault. However, you should not use that column as input for Machine Learning, you could use it to evaluate your findings from ML. All other data columns in a file together somehow indicate if the machine was at fault or not when the data was collected. You need to explore that 'somehow' and Machine Learn that behavior.

Problem 2: (Graduate)

Create an ANN to predict the Alarm column based on the other columns that have pure numeric values. Devise your own data curation process, create network, define activation function, and predict. Even if your network is not fully optimized, please provide your output results with different epoch plotting the loss function. There is no right or wrong network. Grading depends on your steps, approaches and attempts towards making the ANN. Please comment what you have learned about the data. Comment on what ML model would be suitable for this type of dataset.

Hints:

1) There are many columns that has zeros or NAN and there might be some duplicate columns. You may judicially select relevant data columns to perform your machine learning.

2) There is no right or wrong technique for ML. Your effort on ML (supervised or unsupervised or both) is appreciated. Your effort and steps will be graded not the final outcome.

3) Please submit your code with a one-page explanation (in a word file or in a Jupyter Notebook or *.py file with the codes) on your steps taken.

4) Please see the following explanation on the meaning of each column in the data files.

Column Descriptions

Date = Date and Time that sample was taken

Motor coil = Temperature of the motor coil in Celsius

FRONT_IN = Temperature of the Front Inner Bearing

Rear_IN = Temperature of Rear Inner Bearing

FRONT_OUT =Temperature of Front Outer Bearing

REAR_OUT = Temperature of Rear Outer Bearing

MACHINE = Temperature of Machine

VIBRATION = Machine General Vibration

VIB-X = Vibration of “X” Axis

VIB-Y = Vibration of “Y” Axis

VIB-Z = Vibration of “Z” Axis

SPINDLE LOAD = Actual spindle load at time of data collection

SPINDLE COMMAN SPEED = Spindle Speed commanded by Program (M3 S12000)

SPINDLE ACTUAL SPEED = Actual Spindle Speed measured at time of data collection

SPINDLE DIRECTION = The direction spindle is turning STOP,CLOCKWISE,COUNTER CLOCKWISE.

FEED SPEED = Feed commanded by program (inches per minute_

FEED OVERRIDE = Position of the Feed Overide Switch on Operator Panel

FEED ACTUAL SPEED = Actual Feed Speed Measured at time of data sample.

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JOG OVERIDE = Setting of JOG OVERIDE Switch on Operator Panel

FEED AXIS STATUS

NC MODE – machine mode

NC STATUS – machine status

MAIN PROGRAM – program being executed

MAIN COMMENT - comment

MACHINE ALARM – alarm code

MACHINE WARNING – warning code

RELETIVE, ABSOLUTE,MACHINE, DISTANCE, LOAD ALL PERTAIN TO AXIS

1 = AXIS 1 (X AXIS ON MAKINO MACHINE)

2 = AXIS 2 (Y AXIS ON MAKINO MACHINE)

3 – AXIS 3 (Z AXIS ON MAKINO MACHINE)

AXIS 4 THRU 8 DIFFER ON TYPE OF MACHINE. 4 AND 5 NORMALLY CONTROLLABLE WITH HAND WHEEL, BUT 5 – 8 ARE RIDERS. FOR INSTANCE T4'S HAVE TWO BALL SCREWS MOVING THE X AXIS… SO IT WOULD HAVE TWO AXIS FOR X….. AXIS 1 AND EITHER 6,7,OR 8 . THE BEST WAY TO FIGURE OUT AXIS 4 AND 5, WOULD BE TO GO TO EACH TYPE OF MACHINE (T2,T4,MAG3) AND SELECT AXIS 4 ON THE HAND WHEEL AND SEE WHAT AXIS MOVES. THEN THE SAME WITH AXIS 5.

RELETIVE = AXIS RELATIVE POSITION

ABSOLUTE = AXIS ABSOLUTE POSITION

MACHINE = AXIS MACHIEN POSITION

DISTANCE = AXIS DISTANCE TO GO

LOAD = AXIS LOAD

NC MODE = The mode that is selected on the operator panel at time of data sample

NC Status = The status of the NC (Stop / Machining)

SPINDLE POT = Pot number of the tool that is in the spindle

SPINDLE PTN = The Program tool number (PTN) of the tool that is in the spindle.

MAIN PROGRAM = O number of the Main program running on the machine

EXEC PROGRAM = O number of current executing program (could be a sub program called by the main)

SEQ NUMBER = Line number in Program

MAIN COMMENT = Program comment given on same line as ONumber

MACHINE ALARM = Alarm Number of machine alarm

MACHINE WARNING = Machine Warning number

EXEC BLOCK = N Block last read in machine

SPINDLE OVERIDE = Position of Spindle Override Switch on Operator Panel.